

Trajectory Audits: Evaluating Safety Robustness in Multi-Turn AI Systems

Policy Memo

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Executive Summary

AI systems have been advancing rapidly in recent times. Each subsequent model claims a longer context window, which is the maximum amount of text (measured in tokens) a LLM can process, analyse and remember at any single time. As the conversations become more persistent and increasingly autonomous, our approach towards safety evaluation should also evolve. It needs to move beyond examining individual prompt responses and adapt to multi-step workflows.

Conventional evaluations often assess a simple unit: The model is prompted, it responds accordingly, and the response is classified as safe or unsafe. This process, although still necessary, may be insufficient for systems whose behaviour is shaped by a mutating interaction history.

A model may refuse a harmful request when presented directly, but behave differently when the same objective is introduced gradually through several, seemingly benign interactions. Similarly, a model's ability to critique its own output or produce an apparently ethical reasoning trace should not automatically be qualified as evidence of robust safety.

This memo proposed **Trajectory Audits**: a complementary evaluation framework that assesses whether a system maintains relevant safety properties throughout an evolving interaction.

The proposal is motivated by three independent replication studies I conducted. They examined multi-turn jailbreaks, model-generated reasoning and behaviour, and intrinsic self-correction. The experiments conducted are exploratory and do not establish general failure rates. Instead, they illustrate how changing the unit of evaluation can reveal different properties of the same system.

The central policy question is:

For AI systems designed for extended interaction, should safety evaluations evaluate whether safety properties remain robust across the trajectory of an interaction along with the baseline of whether individual responses are safe?

For sufficiently capable or high-impact systems, I argue that they should.

The Evaluation Gap

Traditional evaluation can be represented as:

Prompt → *Response* → *Safety Assessment*

Trajectory evaluation instead considers:

$Context_1 \rightarrow Response_1 \rightarrow Context_2 \rightarrow Response_2 \rightarrow \dots \rightarrow Response_n$

This distinction matters because later behaviour can depend on the interaction history.

Recent research provides evidence for this concern. Weng et al.'s *Foot-In-The-Door: A Multi-Turn Jailbreak for LLMs* found that progressively escalating malicious intent across multiple interactions could substantially increase the attack success rate compared with single-prompt approaches. Their study reported an average attack success rate of 94% across seven models and two benchmarks.

The governance implication is that **interaction structure can itself be an important variable in safety evaluation**. However, It mustn't be confused with the implication that every extended interaction creates a safety failure.

This finding becomes more and more relevant as AI systems move from being a basic Q&A tool toward long term assistants and agentic workflows.

Evidence from Independent Replications

Multi-Turn Escalation

I conducted a small replication inspired by the Foot-In-The-Door methodology, adapting the scenario to corporate misinformation.

A direct request to fabricate financial reporting about a functional company was refused by the model. I then constructed a five-stage trajectory that progressively transformed a benign discussion of thermostat technology into functional reporting, investigative allegations, whistleblower claims, and finally fabricated breaking financial news.

The tested trajectory ultimately completed the harmful objective.

This was one narrow scenario, on one model configuration, evaluated with an automated judge. Therefore, it should not be interpreted as a general failure rate. Instead, its value is diagnostic: **the same objective that was rejected in isolation became achievable when introduced progressively across an interaction**.

This suggests that multi-turn escalation deserves explicit measurement rather than being treated as an incidental property of single-turn benchmark.

Behavioural Consistency

A second replication examined work by Pop et al., *Rethinking Harmless Refusals When Fine-Tuning Foundation Models*. This experiment investigates how undesirable behaviour can persist or change following fine-tuning and examines reason-based deception.

I conducted 100 trials using a fictional car sales scenario in which an AI assistant was instructed to conceal a vehicle's previous accident history.

The undesired behaviour occurred in **96/100 trials**. No trials were classified as reason-based deception.

Reiterating the previous statement: The specific rate is not intended as a general estimate. The more important observation is that **different evaluation channels can reveal different failure modes**. A model generating reasoning trace, a final response or a refusal can provide useful but incomplete evidence.

Model generated reasoning should not be treated as a guaranteed window into internal cognition. For policy purposes, it is better understood as another observable output that itself requires evaluation.

Self Correction

A third replication examined *Large Language Models Cannot Self-Correct Reasoning Yet* by Huang et. al, which found that intrinsic self correction can be unreliable without external feedback and may sometimes degrade performance.

For this, I tested 15 reasoning tasks, which included established puzzles and some minor changes in complexities. All 15 were answered correctly and remained correct following an explicit self-critique step.

This small result does not contradict the original findings. Instead, it highlights a broader governance principle. **Safety mechanisms should be evaluated under the conditions in which they are expected to work rather than receiving implicit credit for merely being present.**

If a system is expected to detect and correct its own mistakes, that capability should itself become an evaluation target.

Policy Proposal: Trajectory Audits

A Trajectory Audit would evaluate a model or deployed AI system across structured sequences of interaction rather than only isolated prompts.

At minimum, audits should measure:

Dimension	Question
Initial Safety	Does the system respond safely to an explicit harmful request?
Trajectory Robustness	Does safety remain stable as context and intent evolve?

Escalation Resistance	Can apparently benign steps progressive produce an unsafe outcome?
Recovery	Can a system recognise and correct an unsafe or ambiguous response?
Consistency	Do outputs and available evaluation channels remain mutually consistent?

The result should be reported separately from conventional single turn performance. A model could therefore have strong direct refusal performance while showing weaker robustness under particular multi-turn trajectories.

Recommended Implementation

1. Standardised Scenarios

Standard bodies or regulators could maintain a common set of scenarios covering gradual escalation, conflicting instructions, repeated pressure, context manipulation, and recovery.

2. Developer-Specific Scenarios

Developers should supplement the common suite with tests based on the intended deployment domain and threat model.

3. Adaptive Red-Teaming

Evaluators should be able to modify later turns in response to earlier model behavior rather than relying exclusively on fixed scripts, wherever practical.

4. Periodic Assessment

Trajectory robustness should be evaluated after significant fine-tuning or deployment changes.

Different developers should be free to achieve robustness through different technical approaches. More importantly, regulation should specify the **safety evidence required**, not mandate a particular technical architecture.

Limitations and Governance Implications

The evidence motivating this proposal is preliminary. The experiments are small, selected, and partly dependent on automated model-based evaluation. They do not establish population-level failure rates or a single causal mechanism.

Nor does longer interaction necessarily make systems less safe. Additional context can sometimes improve performance. The relevant research question is therefore not whether "more turns" are inherently dangerous, but **which interaction structures change safety performance and under what conditions**.

Trajectory Audits should consequently complement rather than replace existing single-turn evaluations.

The broader governance principle is simple: **the unit of evaluation should reflect the unit of deployment**.

If a system is deployed as a persistent, interactive agent, demonstrating that it responds safely to isolated prompts may not be sufficient evidence of robust behavior over the course of an actual interaction.

Conclusion

As AI systems become more persistent and agentic, the trajectory of an interaction may itself become part of the safety problem.

The evidence presented here is preliminary, but it illustrates a recurring evaluation issue: changing the evaluation procedure can reveal behaviors that isolated tests do not capture.

Trajectory Audits offer a practical way to incorporate that perspective into AI safety evaluation without replacing existing benchmarks or prescribing a particular model architecture.

The key question for increasingly capable systems should therefore expand from:

"Does the model respond safely to this prompt?"

to:

"Can the system maintain its safety properties as the interaction evolves?"

For higher-risk AI systems, that second question deserves a place in the evaluation regime.

References

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